Blepharitis

Lorne Yudcovitch, OD, MS, FAAO

SOAP Format

- **S = Subjective**
  - **AKA symptoms**
  - **Everything the patient says**
- **O = Objective**
  - **AKA signs**
  - **All the doctors observes and all the test results**
- **A = Assessment**
  - **AKA diagnosis (or diagnoses)**
  - **What is wrong with the patient**
- **P = Plan**
  - **For each diagnosis, what is going to be done to help the patient**

Blepharitis

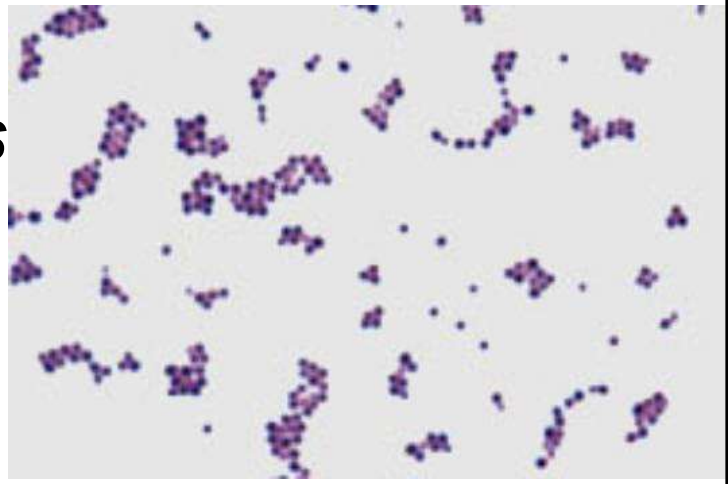
- Anterior blepharitis - inflammation centered around eyelashes and follicles
  - Staphylococcal
  - Seborrheic
  - Mixed
- Posterior blepharitis - inflammation involving meibomian glands and gland orifices
  - Meibomian seborrheic
  - Meibomian gland dysfunction (AKA meibomianitis, meibomitis)
- Angular blepharitis
  - Staphylococcal
  - Moraxella

Staphylococcal Blepharitis

Etiology

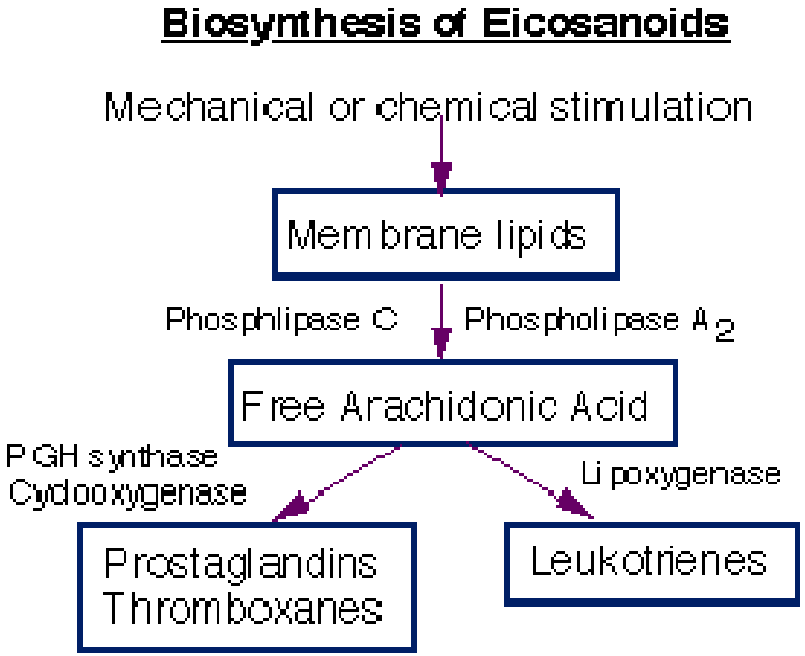
Infection of the lash follicles and lid margins

- **With *Staphylococcus aureus* and/or**
  - ***S. aureus* is normally present on lid margins of small number of individuals**
- **overgrowth of *Staphylococcus epidermidis***
  - ***S. epidermidis* normally present on lid margins (normal flora) of most individuals**
- **→ inflammatory response by the body**



Staphylococcal Blepharitis

- **Secondary mechanism of inflammation**
  - **Production by the bacteria of lipase**
  - **→ breaks down the lipids in the tears**
  - **→ free fatty acids (FFAs)**
  - **→ trigger inflammatory and irritative response**



Staphylococcal Blepharitis

Inflammatory response:

- **Dilation of blood vessels (redness)**
- **Increased vascular permeability (edema + blood proteins)**
- **Movement of PMNs from blood into tissue**
- **If inflammatory response is caused by bacteria, lots of PMNs move from blood into tissues to fight infection through phagocytosis**
- **→ dead PMNs + protein clot = pus**



- **<u>**** Presence of pus = bacterial infection</u>**

PMNs = polymorphonucleocytes (white blood cells)

Staphylococcal Blepharitis

Subjective

- **Depends on severity**
- **Chronic irritation and burning**
- **FB (foreign body sensation)**
- **Tender lids (may have recurrent hordeola or chalazia)**
- **Patient may notice crusts on lashes**
- **The lids may be stuck together in morning**
- **Patient may complain of red, irritated lids**

Staphylococcal Blepharitis

Objective

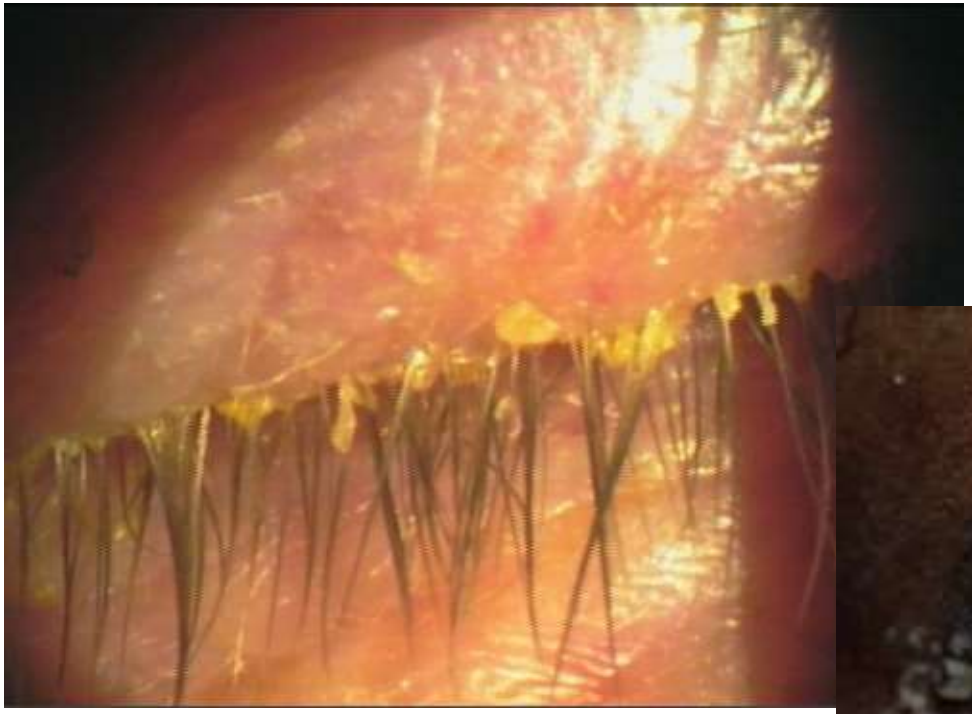
- **Crusts along the lash line at the base of the lashes**
- **Dried pus**





Staphylococcal Blepharitis

More crusts





Staphylococcal Blepharitis





Staphylococcal Blepharitis

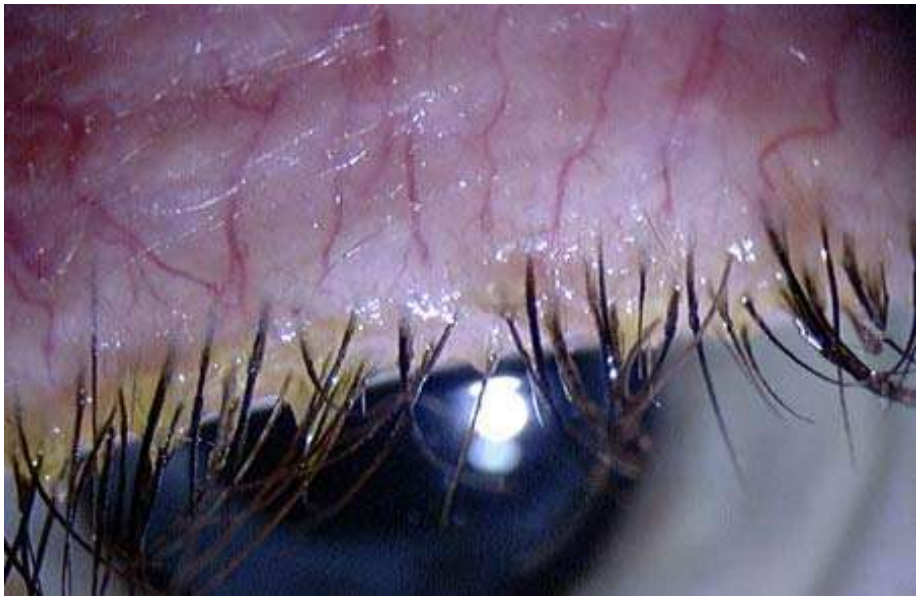
- **Crust can encircle lash = collarette**



Staphylococcal Blepharitis

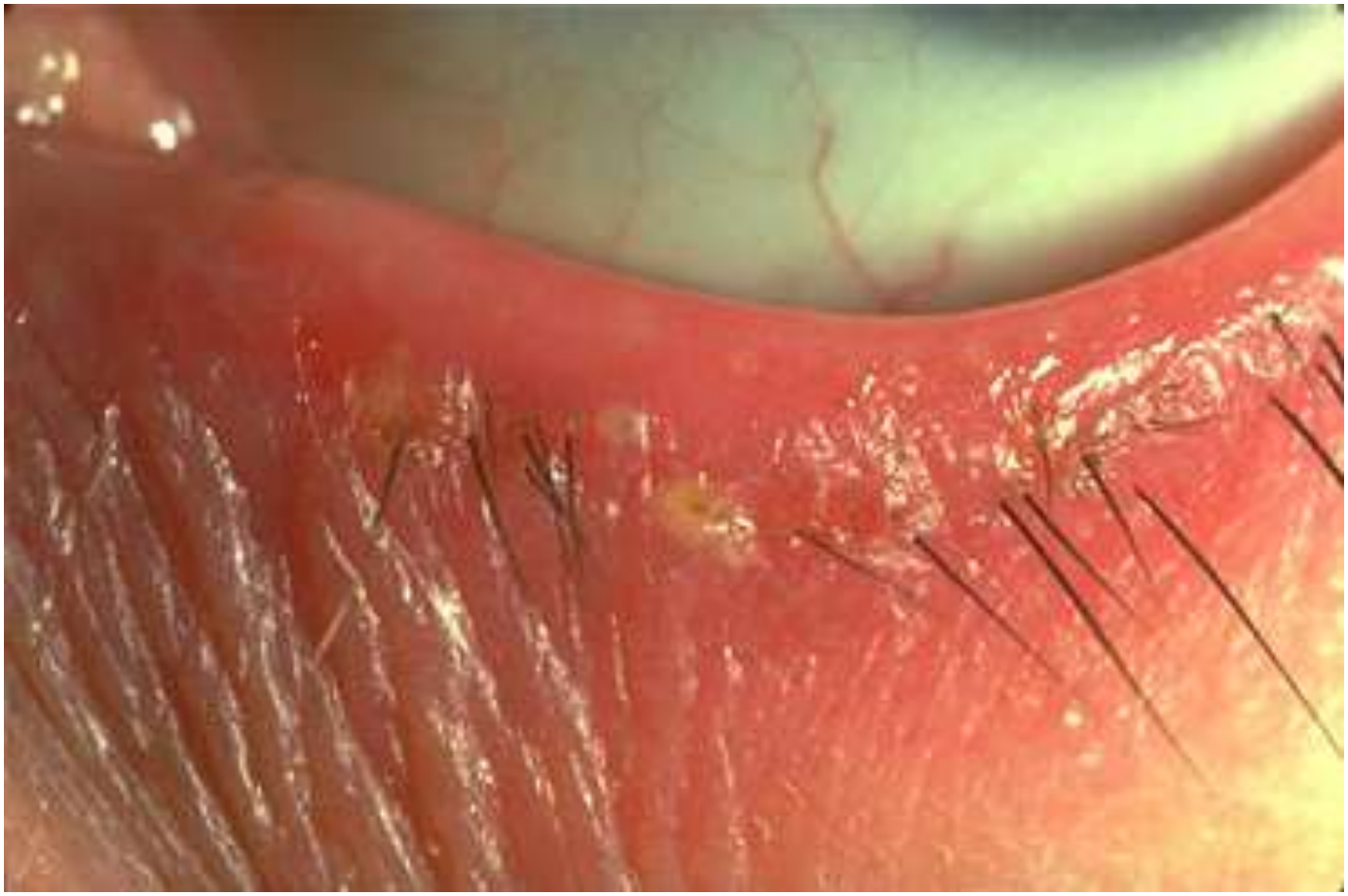
- Crust at the lash base – can also appear ‘oily’





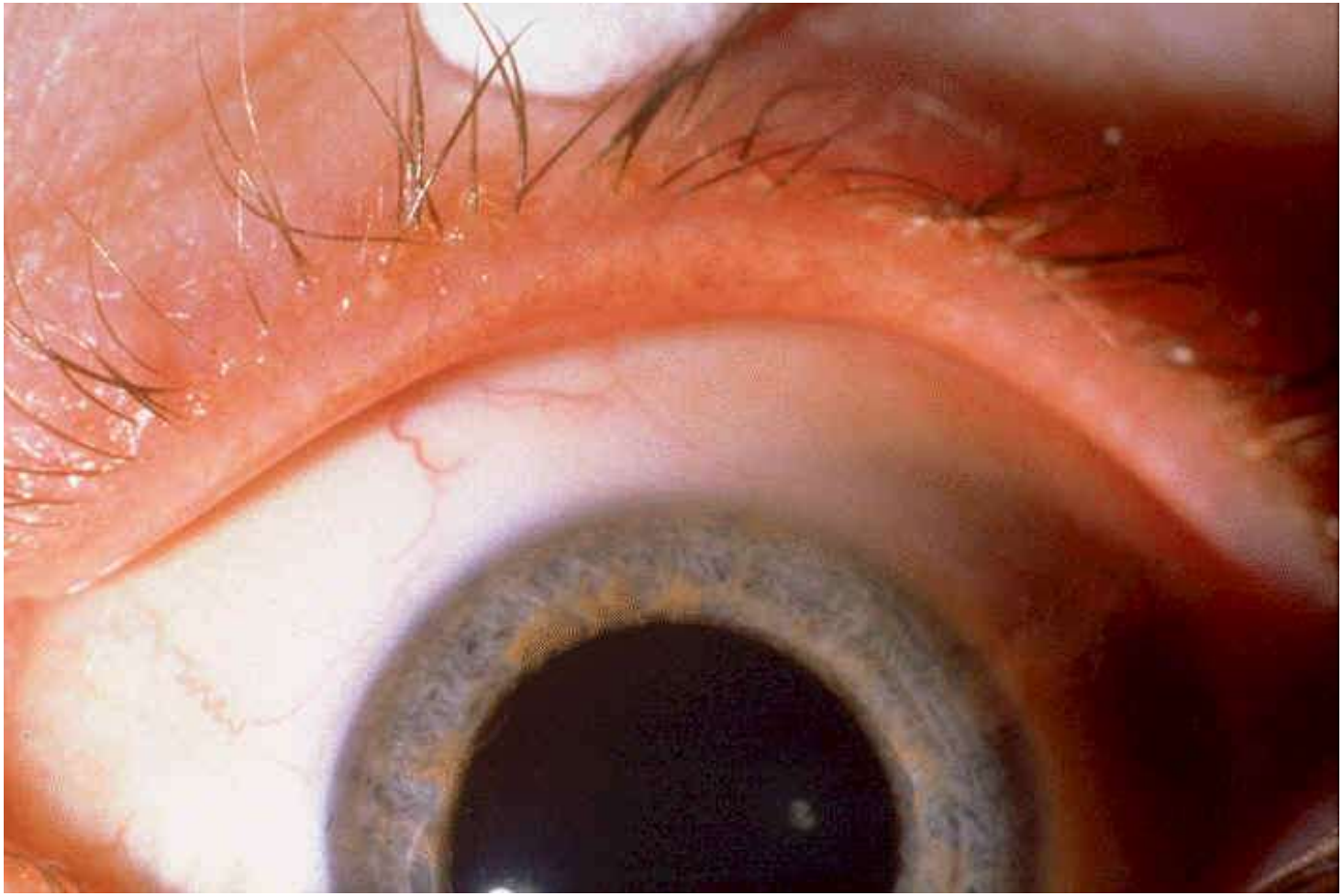
Staphylococcal Blepharitis

- **Infected follicles and madarosis (loss of lashes)**



Staphylococcal Blepharitis

- **Infected follicles**



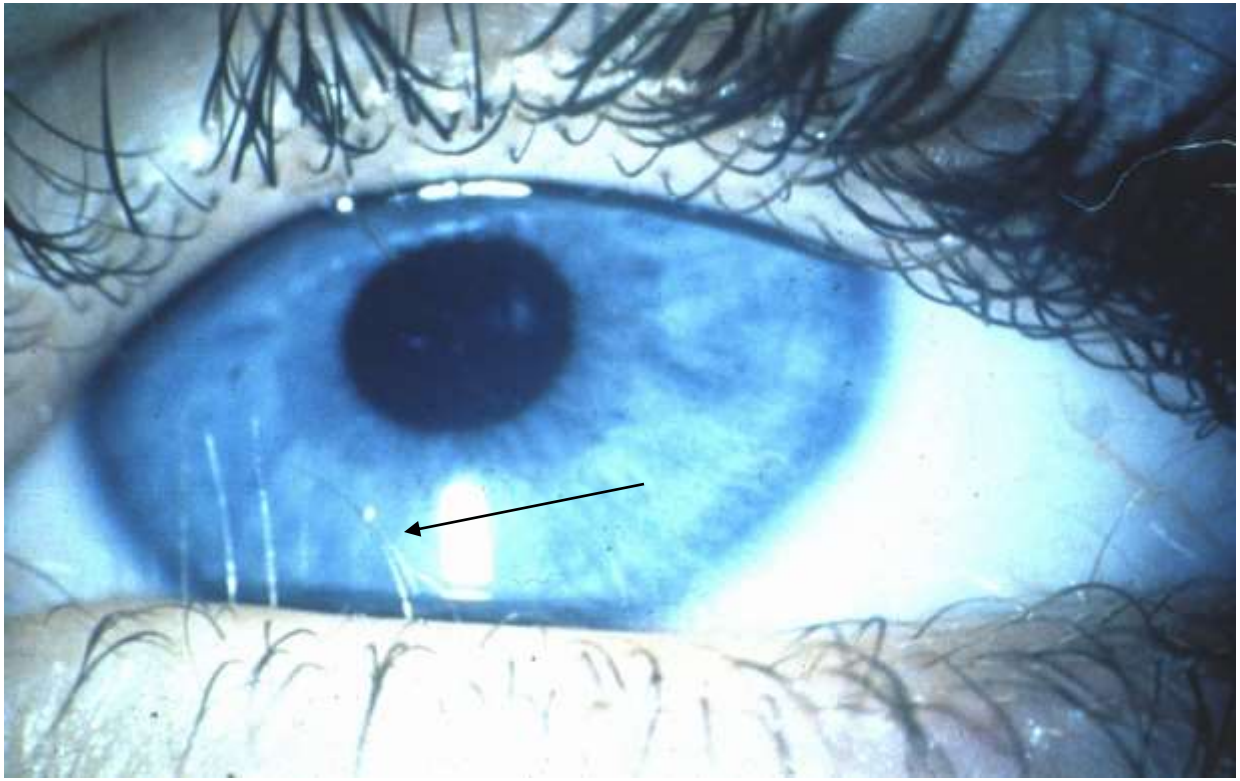
Staphylococcal Blepharitis

- **Ulcers at the base of the lashes**



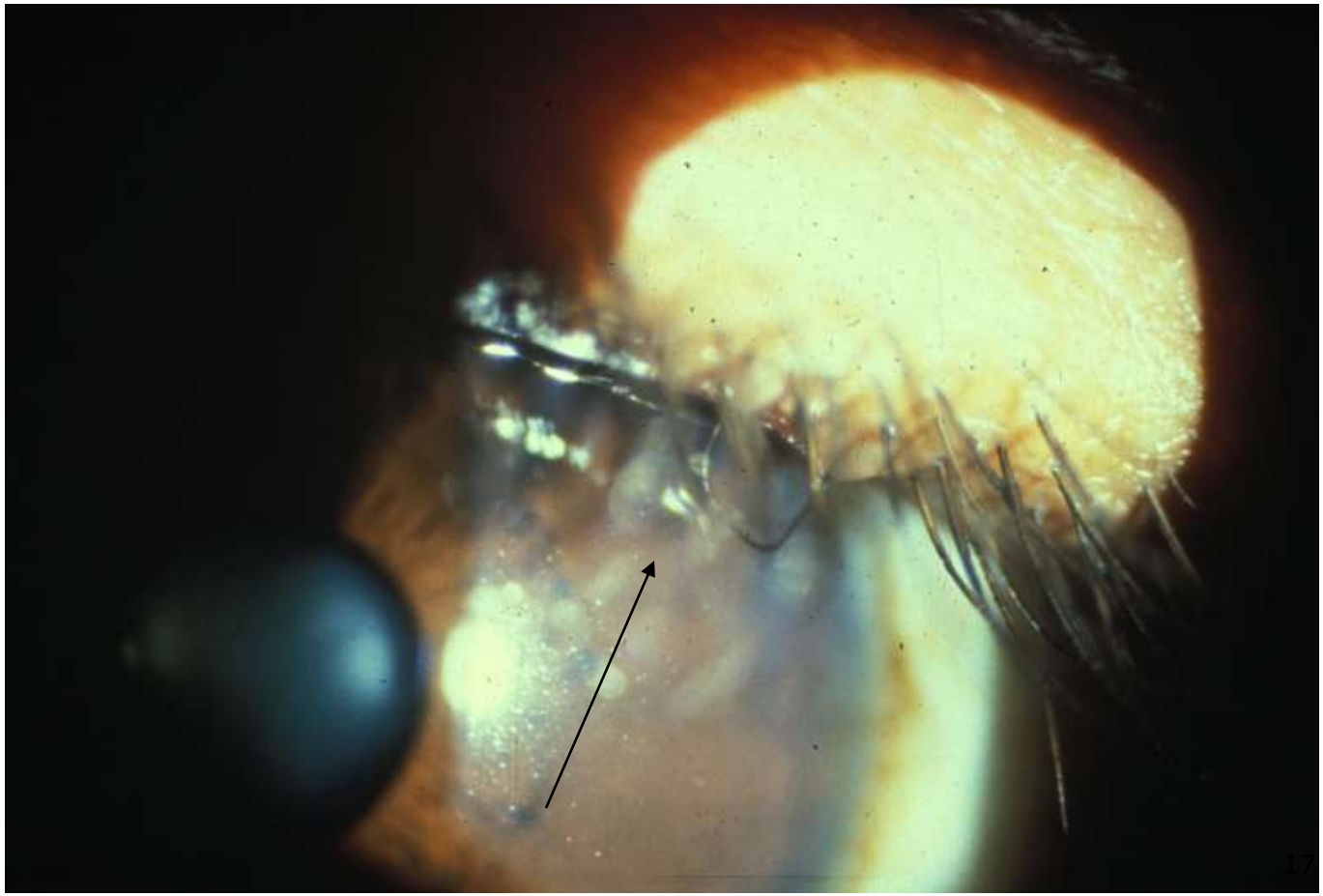
Staphylococcal Blepharitis

- **Damage to the lash follicles → trichiasis (lashes that point towards the cornea and may actually scratch the cornea)**



Staphylococcal Blepharitis

- **Trichiasis - lashes need to be epilated if touching cornea**



Staphylococcal Blepharitis

- Red and thickened (tylosis) lid margins
  - Seen mostly with chronic disease
  - Also note madarosis and conjunctivitis
- Lid margins may become indurated (hardened) over time



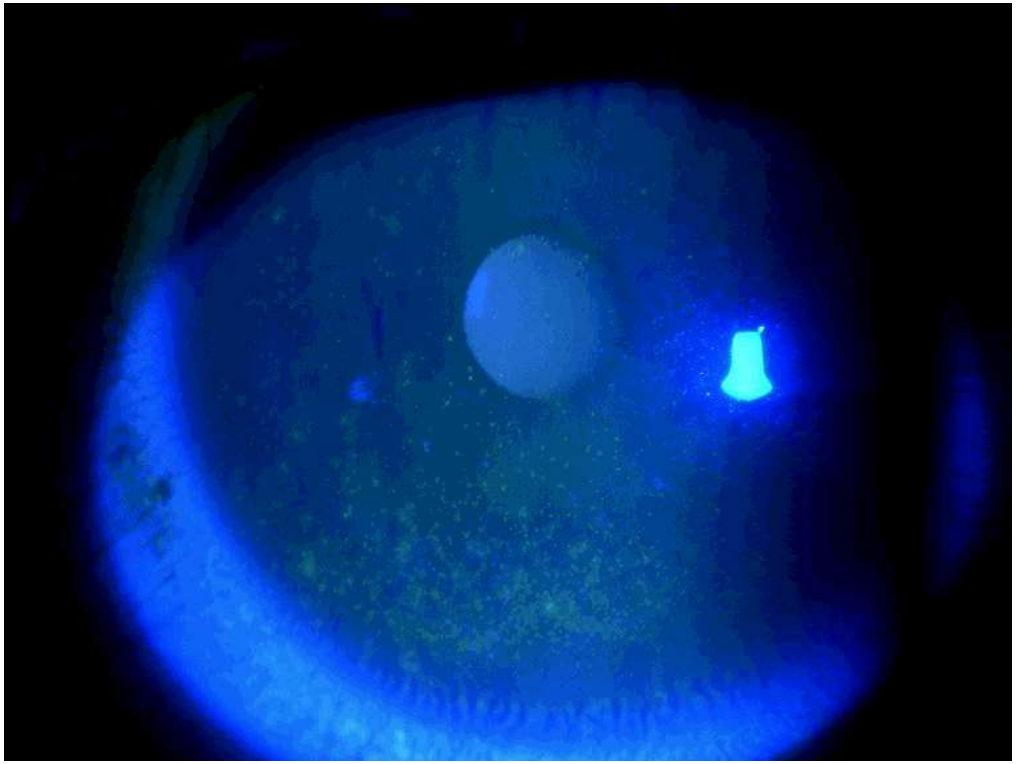
Staphylococcal Blepharitis

- **May have secondary conjunctivitis (usually papillary)**
  - **Sometimes called staphylococcal blepharoconjunctivitis**



Staphylococcal Blepharitis

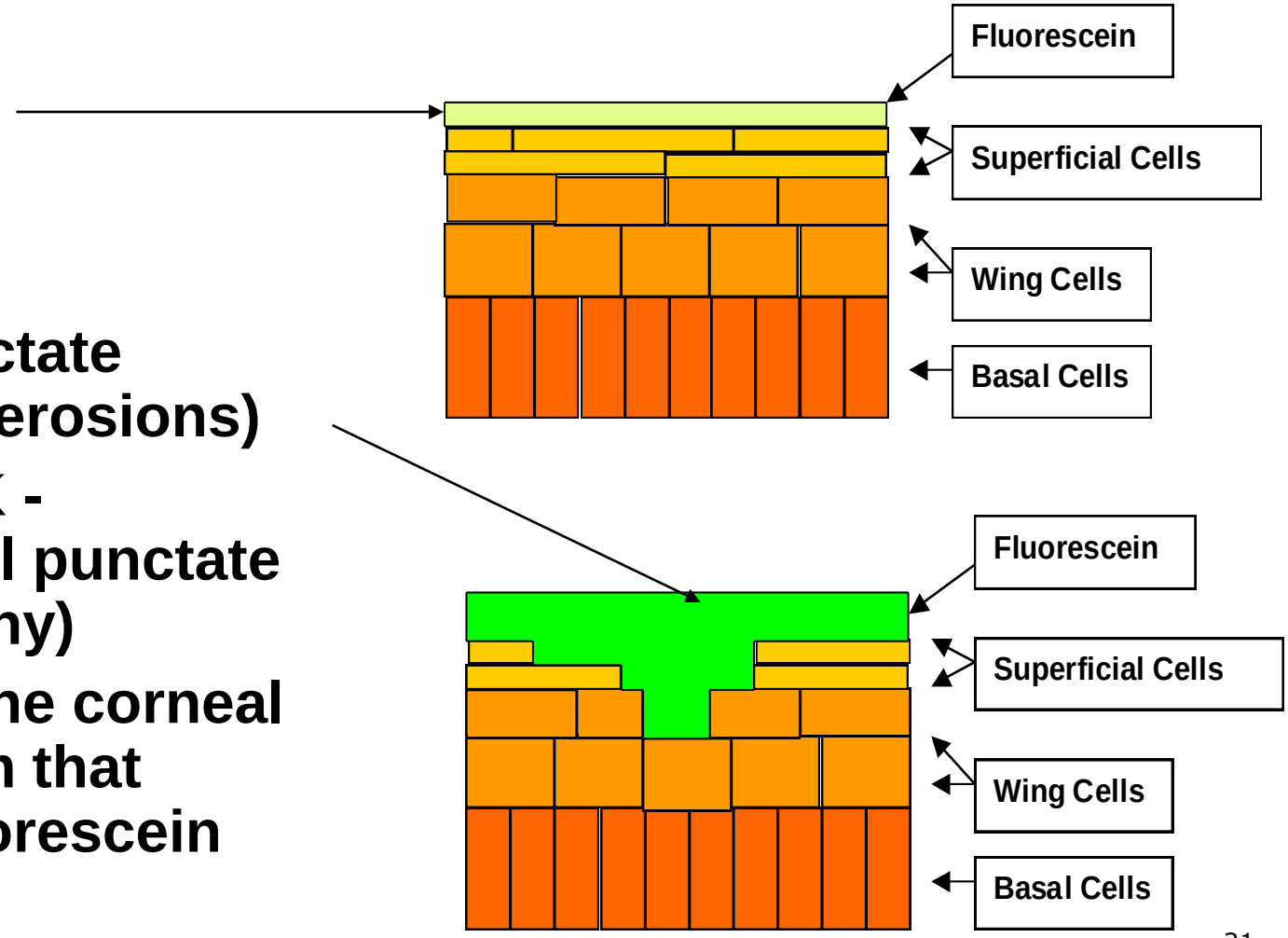
- Punctate epithelial erosions (PEE)
  - Result from reaction to bacterial exotoxins
- PEE in inferior third of the cornea
- PEE stains with fluorescein (green color with colbalt blue filter)



Staphylococcal Blepharitis

- Normal

- PEE (punctate epithelial erosions)
- (AKA SPK - superficial punctate keratopathy)
- Divits in the corneal epithelium that cause fluorescein to fluress



Staphylococcal Blepharitis

- **Diagnosis**
  - **Usually based on clinical findings**
  - **Severity will dictate appropriate treatment**

Staphylococcal Blepharitis

Treatment - lid hygiene

- **Remove crusts**
- **Remove bacteria**
- **Increase blood circulation to lid margin**
- **Two step process - warm compresses and lid scrubs**



Staphylococcal Blepharitis

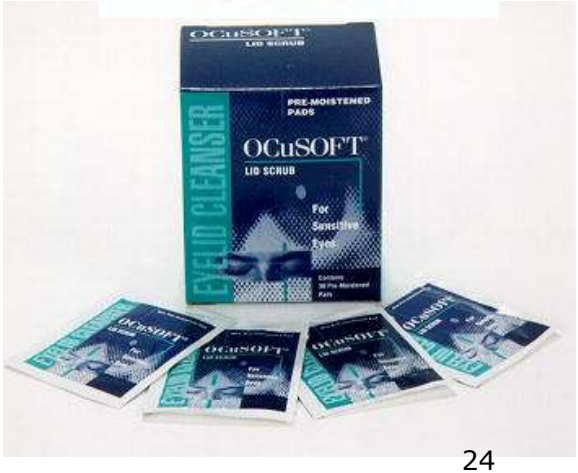
Warm compresses

- Soak lids with warm, wet compress (eyelids closed)
- 10-15 min bid-qid



Lid scrubs

- Diluted baby shampoo and water or
- Commercially available cleansers
- Scrub lid margins
- Bid to qid after warm compresses
- Hypochlorous acid 0.02%



Staphylococcal Blepharitis

Baby Shampoo – the 'old' treatment modality

- **The baby shampoo is first diluted one-to-one with water in 'cup' in palm of hand.**
- **Then mixed by rubbing with clean fingertips and applied in gentle oval scrubbing motion to margin and eyelash base of closed eyelid for 1 minute, followed by fresh water facial rinse.**
- **Baby shampoo can alternatively be diluted in container (e.g. plastic cup) and scrubbing performed using washcloth wrapped around finger (after dipping it in the diluted shampoo). A cotton tip applicator may be used alternatively.**
- **NOT recommended anymore due to drying effects**



Hypochlorous acid 0.02%

- Ocusoft HypoChlor™
- Spray or gel
- Used as needed or daily
- Can be used with lid wipes
- Spray form often used AM
- Gel form often used PM
- Generic versions as well



Staphylococcal Blepharitis

Antibiotic ointment applied to lid margins

- **Bacitracin**
  - **Cell wall inhibitor**
- **Erythromycin**
  - **Macrolide**
- **Gentamicin**
  - **Aminoglycoside**
- **Tobramycin**
  - **Aminoglycoside**



Regimen

- **Moderate: qhs or bid**
- **Severe: tid or qid**

Prescription Writing

Christina Gosses, OD

354 NW Orange Lane

Chicago, IL 60631

Name Sherman Pierce Age 22 Date 09/22/24

Address 2043 College Way, Forest Grove OR 97116

℞

Bacitracin ophthalmic ung
Dispense: one 3.5 gram tube
Instructions: apply 1/4 inch to each lid margin of both eyes twice a day for 10 days

X No Refill _____,O.D. _____,O.D.

☐Refill x ___ Dispense as Written Product Selection Permitted

Staphylococcal Blepharitis

Lubricants

- **Dry eye secondary to lid condition**
- **Artificial tears qid or prn**

Oral Antibiotics

- **Valuable when unusually severe**
- **Doxycycline: 100mg PO bid x 7-10 days**
  - **Tetracyclines inhibit lipase synthesis by bacteria**
  - **Doycycline inhibits Staphylococcal exotoxin-induced cytokines and chemokines**
  - **Are helpful even if bacteria are resistant to antibiotic effect**

Staphylococcal Blepharitis

Follow-up

- **1-3 weeks depending on severity and medications prescribed**
- **Consider follow-up 2-3 weeks later**
- **If cornea involved, follow-up reduced to days**

Patient education

- **Chronic condition - Usually exacerbations and remissions**
- **Goal: to control not cure**
- **Importance of daily lid hygiene**
- **Medication for acute flare-ups**

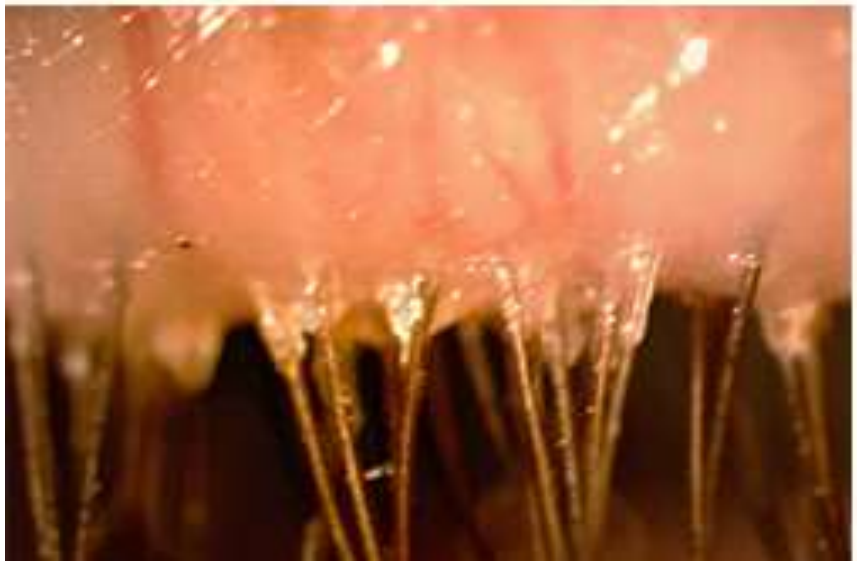
BlephEx – Blepharitis treatment option

- Rysurg
- Medical grade disposable micro-sponge spins along edge of eyelids and lashes
- Removes scurf/debris and exfoliates eyelids
- Proparacaine usually instilled prior to treatment
- Eyes rinsed afterwards
- 6 -8 minute procedure

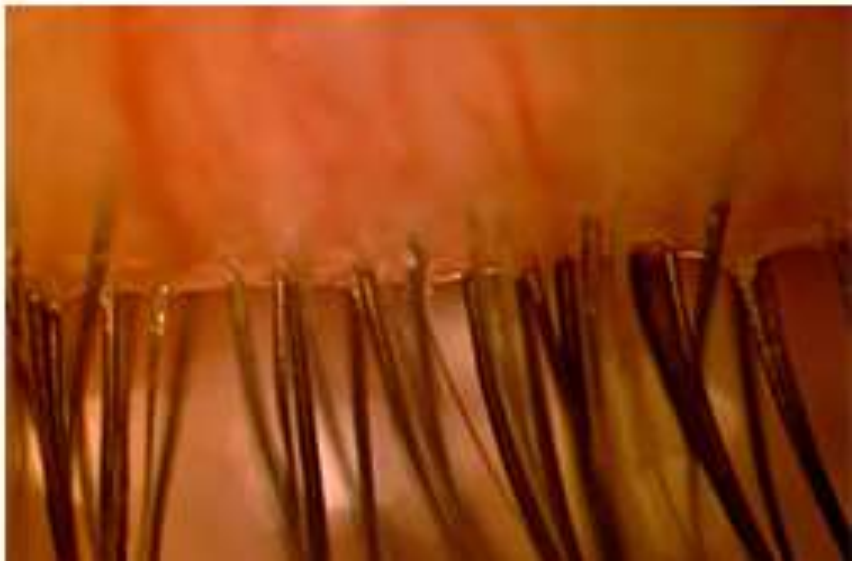


Blephex

Before



After



Staphylococcal Blepharitis Associated Marginal Infiltrate

- White blood cells infiltrate cornea between epithelium and stroma (subepithelial)
  - White spot in corneal periphery with clear area between it and limbus
  - Does not stain with fluorescein
- When the white cells die, they release enzymes that can damage the epithelium above the infiltrate
  - The infiltrate may then stain in a sporadic punctate pattern



Staphylococcal Blepharitis Associated Marginal Infiltrate

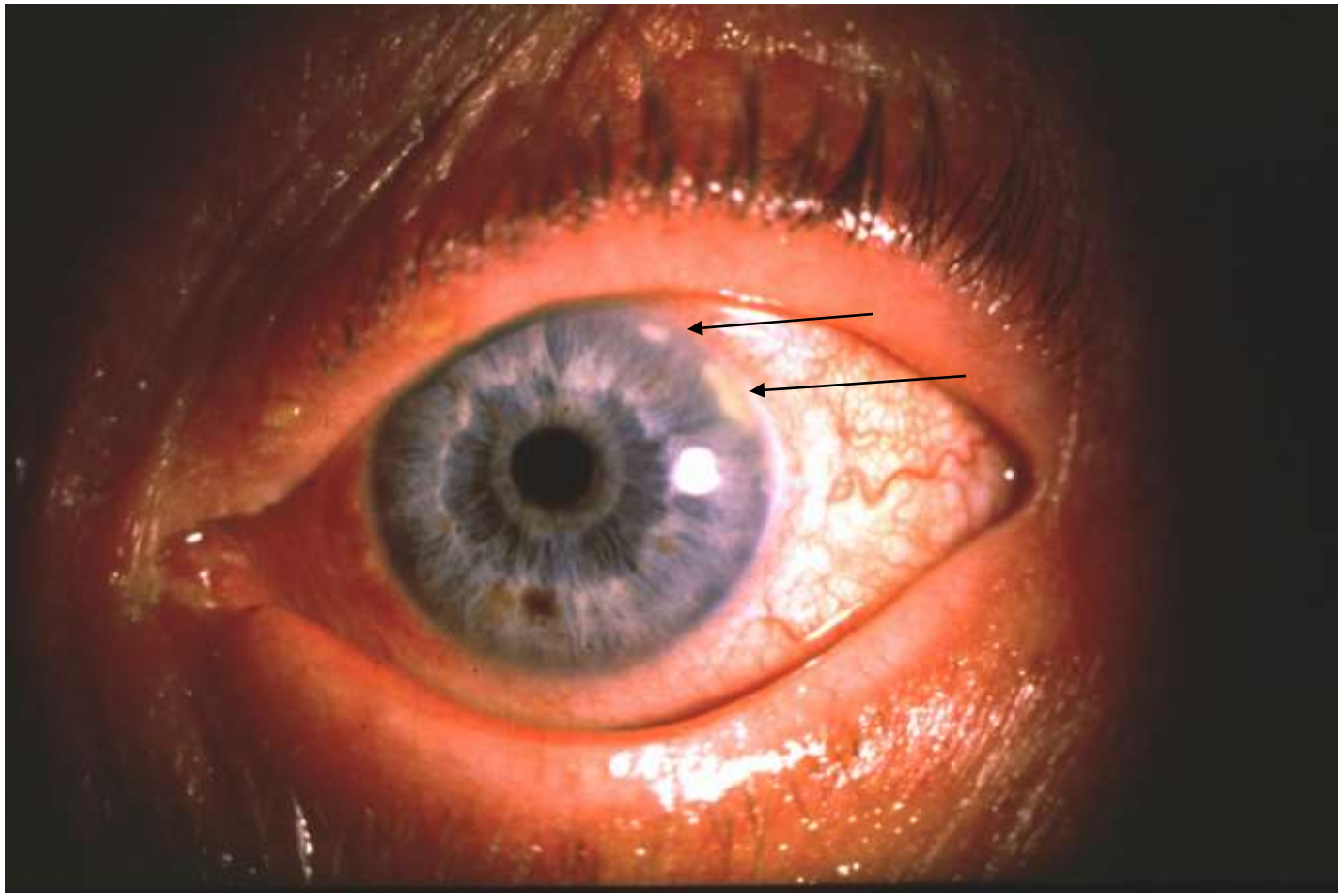
Hypersensitivity to staphylococcal toxins

Ag-Ab-Complement triggered

- **Ag (staph product) +**
- **Ab (produced in conjunctiva) +**
- **complement components (from limbal blood vessels) →**
- **C3a and C5a (chemotactic factors for white cells from the limbal blood vessels) →**
- **White spot in cornea of an accumulation of white cells**

Staphylococcal Blepharitis Associated Marginal Infiltrate

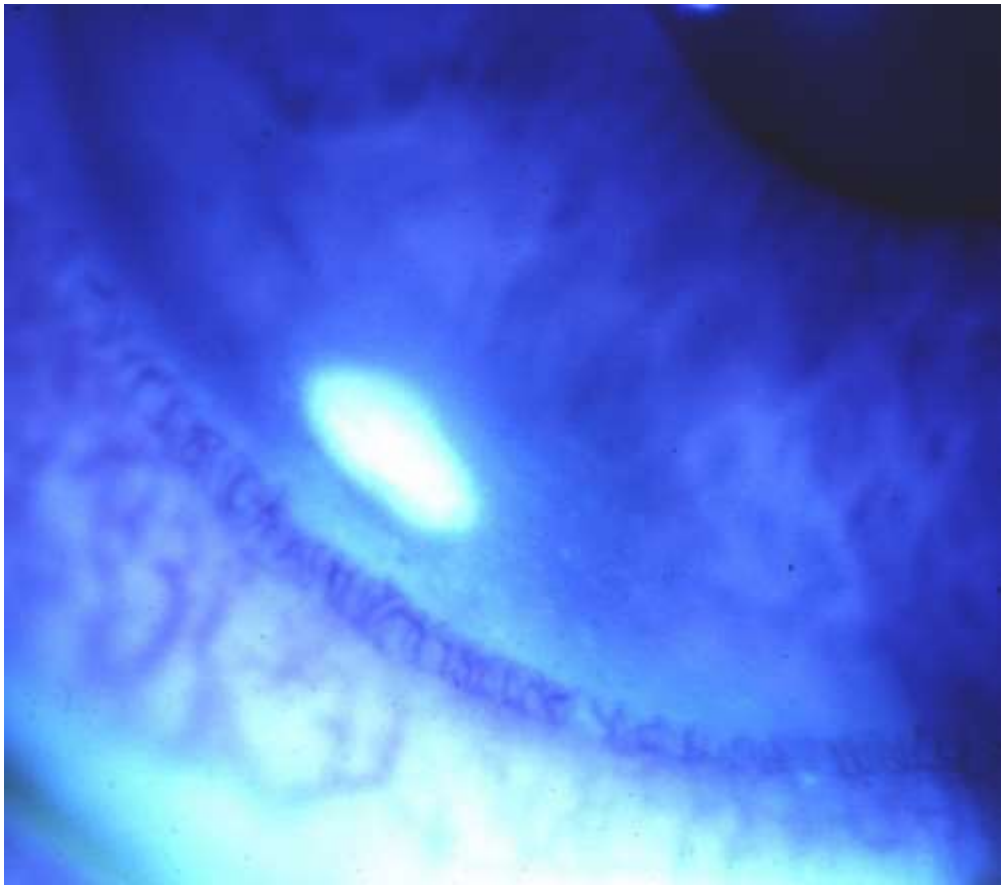
- **Marginal infiltrates**



Staphylococcal Blepharitis Associated Marginal Infiltrate

A marginal infiltrate that does not stain





Staphylococcal Blepharitis Associated Marginal Infiltrate

Treatment

- **Treat the staphylococcal blepharitis and add:**

- **Topical steroid q2h to qid**
  - **Prednisolone acetate**
  - **Dexamethasone**
  - **Loteprednol**

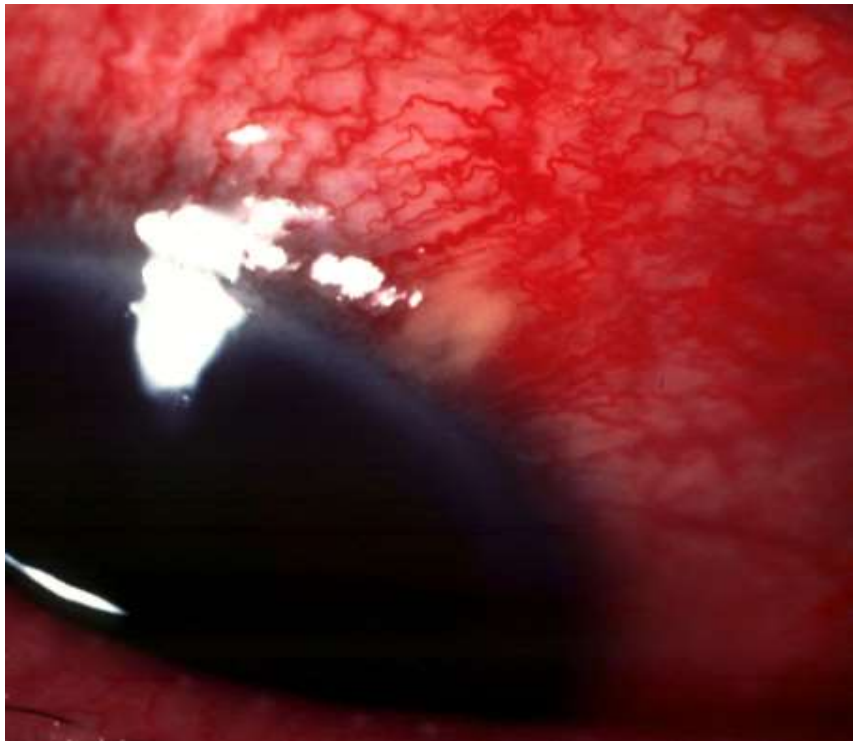
Or…



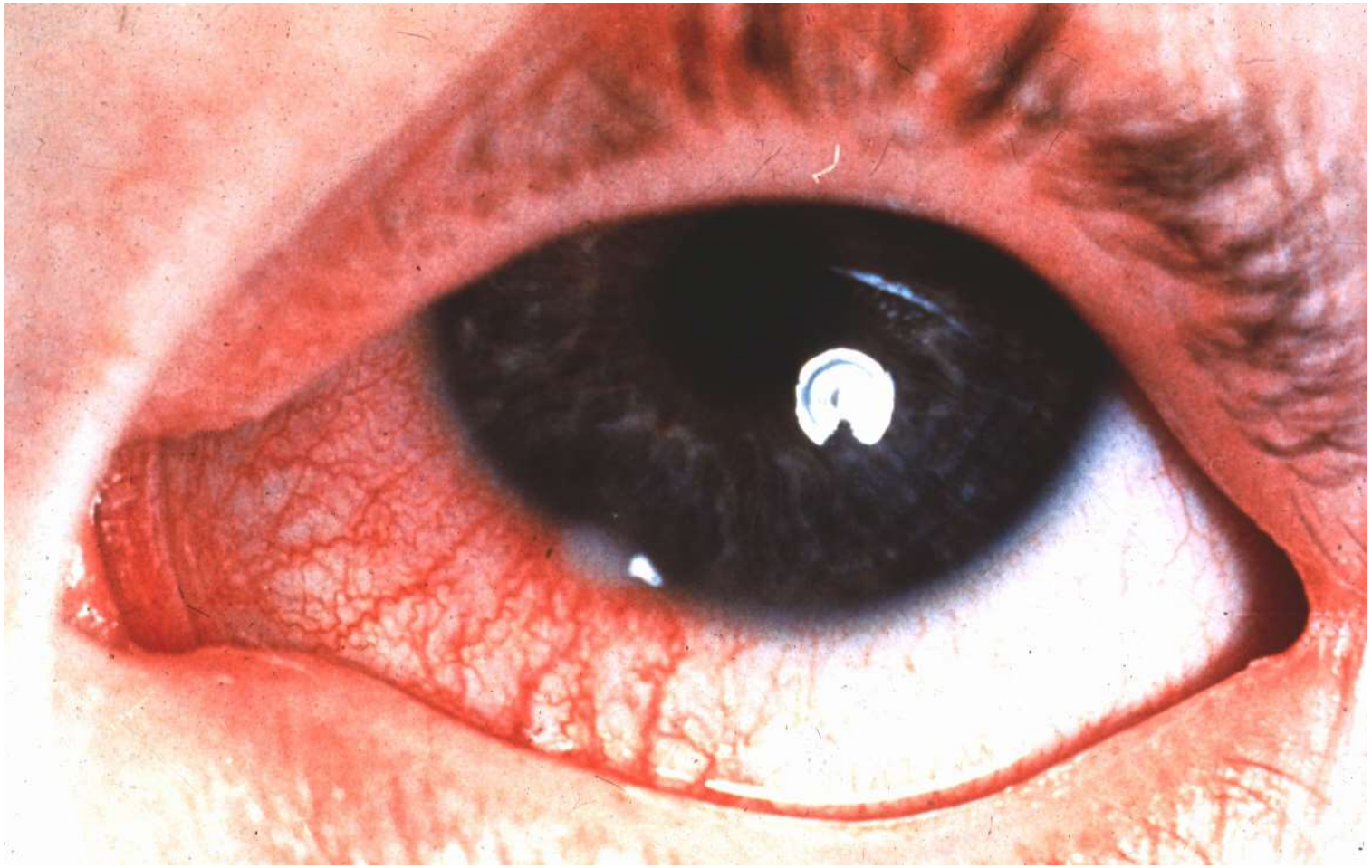
- **Topical antibiotic/steroid combination such as tobramycin with dexamethasone (Tobradex), or tobramycin with loteprednol (Zylet) q2h to qid**

Staphylococcal Blepharitis Associated Phlyctenule

- Hypersensitivity reaction to staph toxins
- Type IV reaction
- Raised white lesion on conjunctiva, limbus, or cornea
- Can move over cornea, pulling blood vessels and scarring as it goes



Staphylococcal Blepharitis Associated Phlyctenule



Staph Blepharitis Associated Phlyctenule

Treatment

- **Treat the staphylococcal blepharitis and add:**



- **Topical steroid q2h to qid**
  - **Prednisolone acetate**
  - **Dexamethasone**
  - **Loteprednol**

Or…

- **Topical antibiotic/steroid combination such as tobramycin with dexamethasone, or tobramycin with loteprednol q2h to qid**

Staph Blepharitis Associated Phlyctenule

- Note – several years ago a primary cause of phlyctenule was tuberculosis (TB)
- Patients who were:
  - Recent immigrants from areas endemic with TB, or
  - Individuals who acquired active TB locally
- Now, Staphyloccus has once again become a primary cause of phlyctenule

Seborrheic Blepharitis

Etiology

- **Unknown**
- **Often associated with seborrheic dermatitis**
  - **Chronic disorder of the skin/head areas where sebaceous glands are abundant**
  - **Dandruff (AKA scurf) - excessive shedding of dead skin cells of the scalp - is often a symptom of seborrheic dermatitis**



Seborrheic Blepharitis

Subjective

- **Frequently asymptomatic**
- **Possible morning “mattering”**
- **Dry eye complaints (poor tear film)**
- **Symptoms may depend on other concurrent processes**

Seborrheic Blepharitis

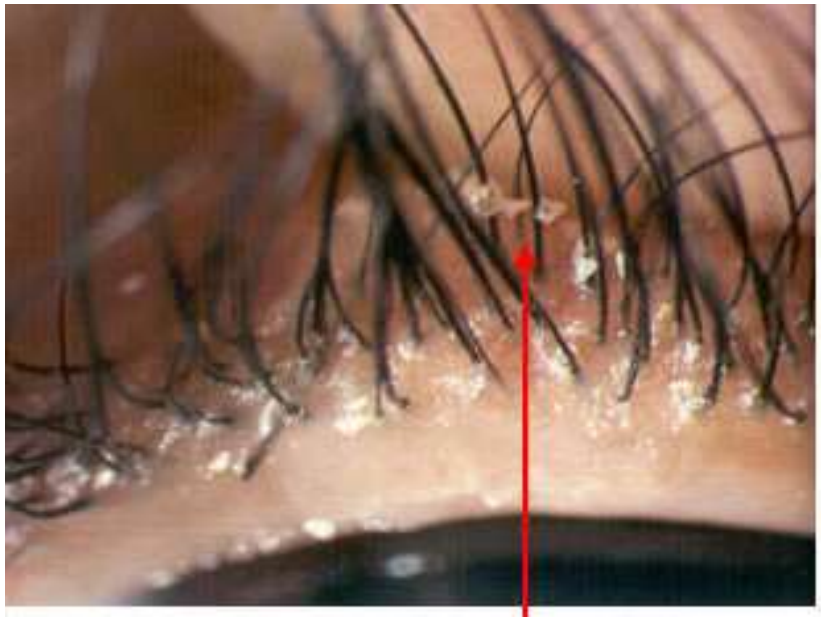
Objective

- **Greasy, waxy scales at the base of the lashes that flake off easily (dandruff-like)(AKA scurf)**
- **Skin of the lid a little greasy**



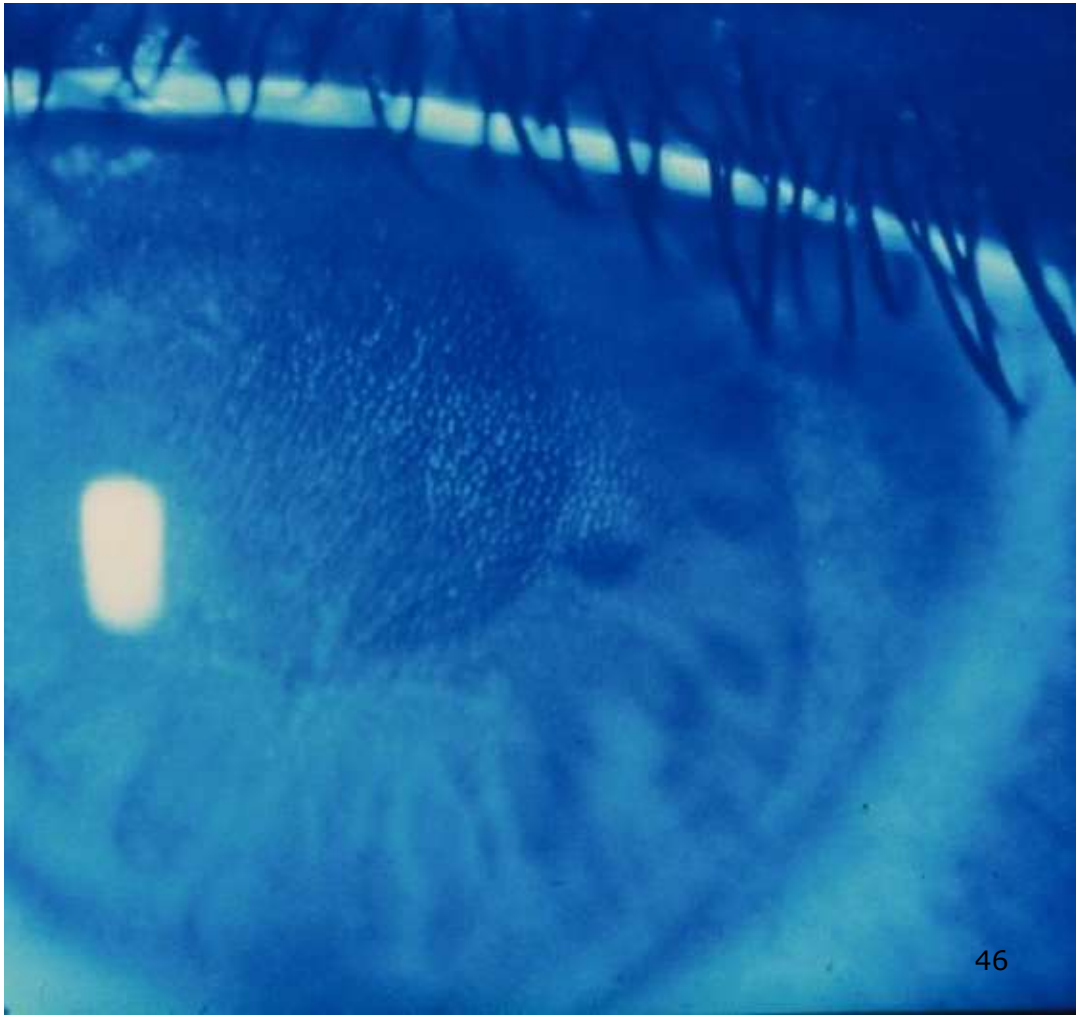
Seborrheic Blepharitis





Seborrheic Blepharitis

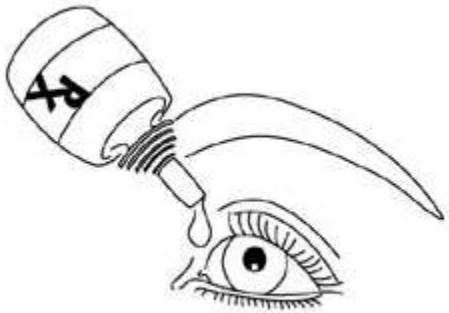
- **Interpalpebral PEE**



Seborrheic Blepharitis

Treatment:

- **Lid hygiene**
- **Lubricants**
- **Follow-up 2-4 weeks**
- **Patient education on chronicity**
  - **Lid hygiene daily**





Mixed Blepharitis

- **Seborrheic blepharitis with associated staphylococcal superinfection**
- **Plan**
  - **Same as for staph blepharitis**

Meibomian Seborrheic Blepharitis

Etiology

- **Unknown**
- **Excessive oil production**
- **Quality of oil is abnormal**

Subjective

- **Dry eye complaints**

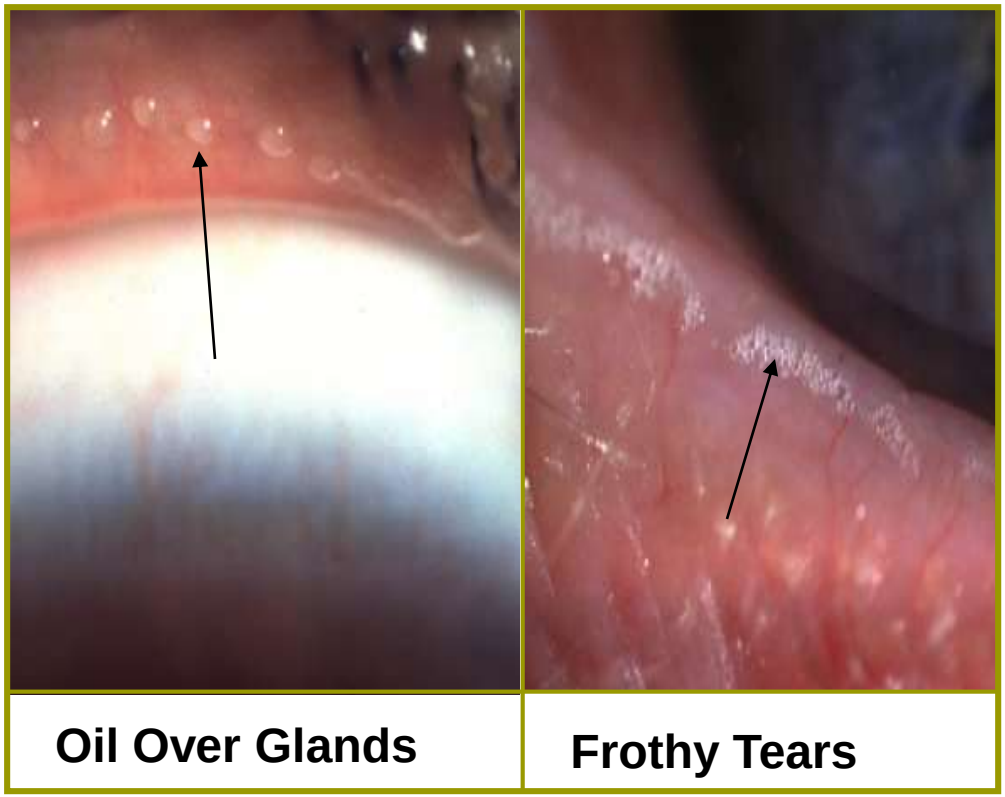
Meibomian Seborrheic Blepharitis

Objective

- Frothing of tears
- "Oil slick" in tear film
- TBUT may be reduced
- Conjunctival injection
- Lid tissue appears normal

Treatment

- Keep lid margins clean
- Lubricants



Oil Over Glands | Frothy Tears

Angular Blepharitis

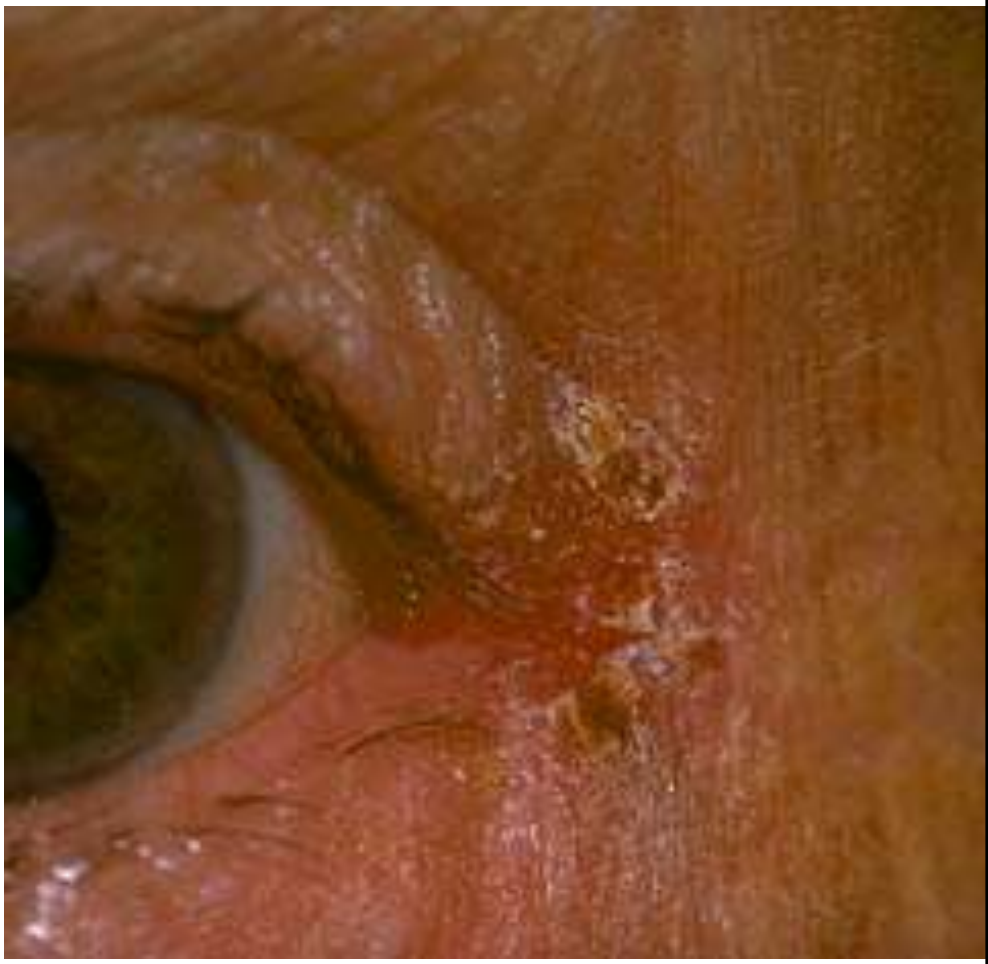
- Localized eczematoid inflammation of lid at outer canthus and sometimes medial canthal region



- Staphylococcal form:
  - Dry and scaly
- Moraxella form:
  - Wet
  - Macerated lid with whitish, frothy discharge
- Angular blepharitis may involve conjunctiva
- All forms of angular blepharitis call for treatment with antibiotic ointment
  - May need to try different ABs based on bacteria type

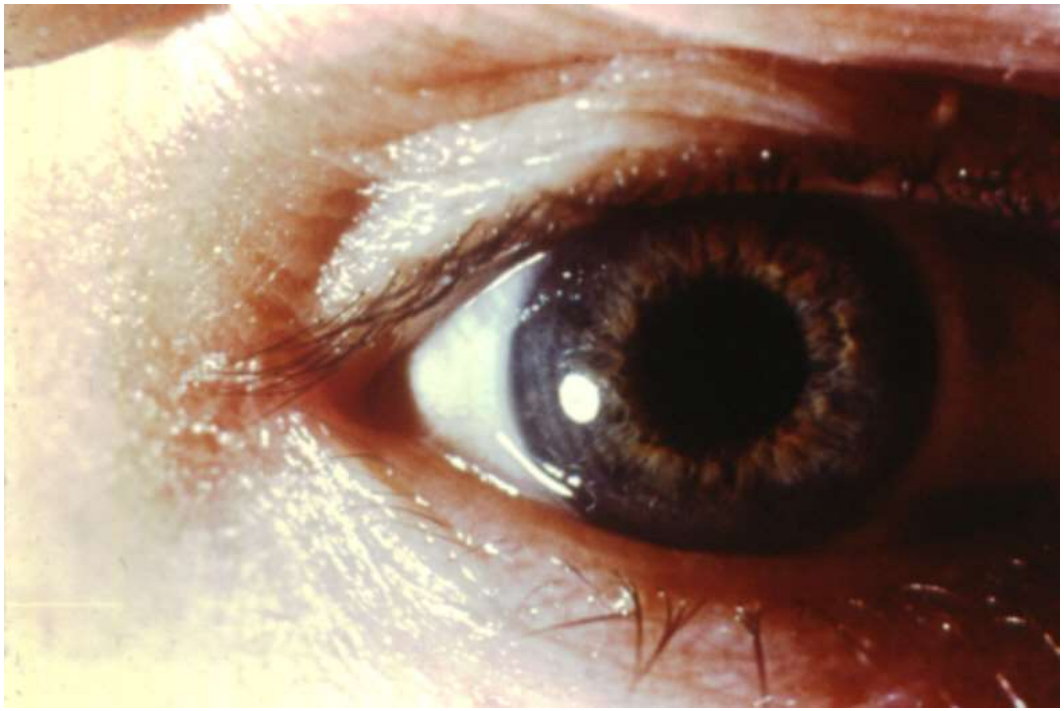
Angular Blepharitis

- Chronic lid irritation
- Skin drying, scaling, cracking in outer canthi
- Cause
  - *Moraxella lacunata*
  - *Staphylococcus epidermidis*
  - *Staphylococcus aureus*



Angular Blepharitis

- ***Moraxella*: zinc sulfate 0.25% solution**
- ***Staphylococcus*: bacitracin or erythromycin**



THANK YOU!